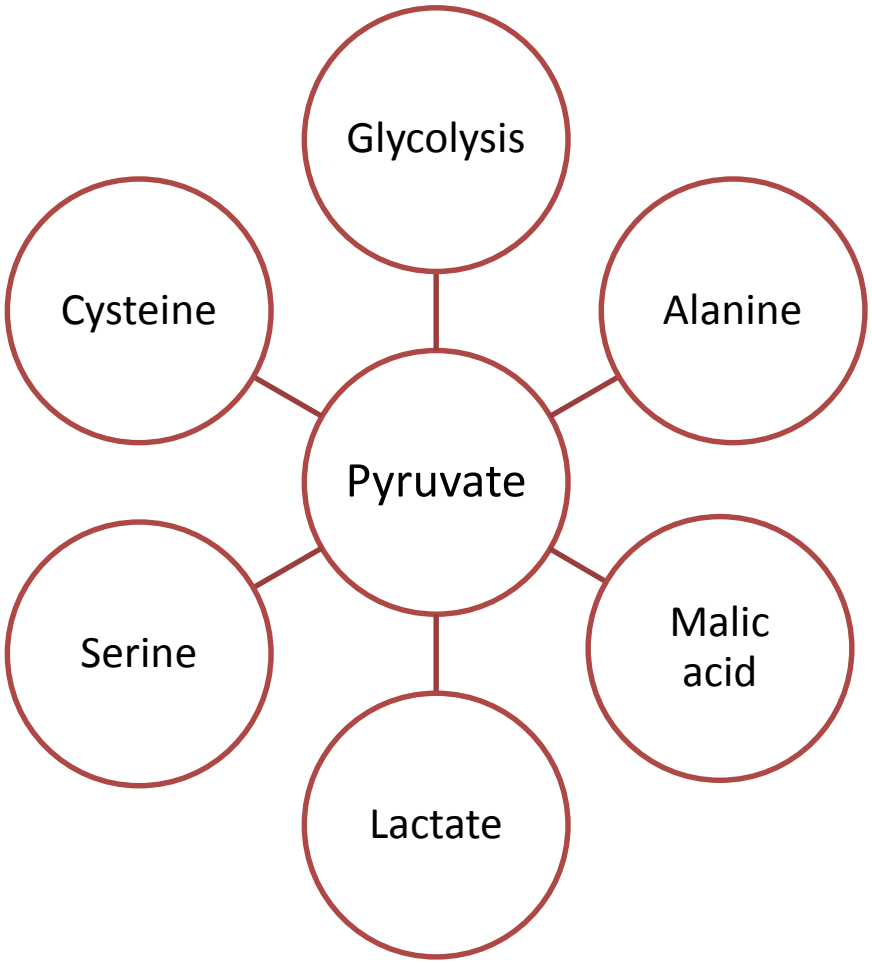
Fate of Pyruvate- A quick review

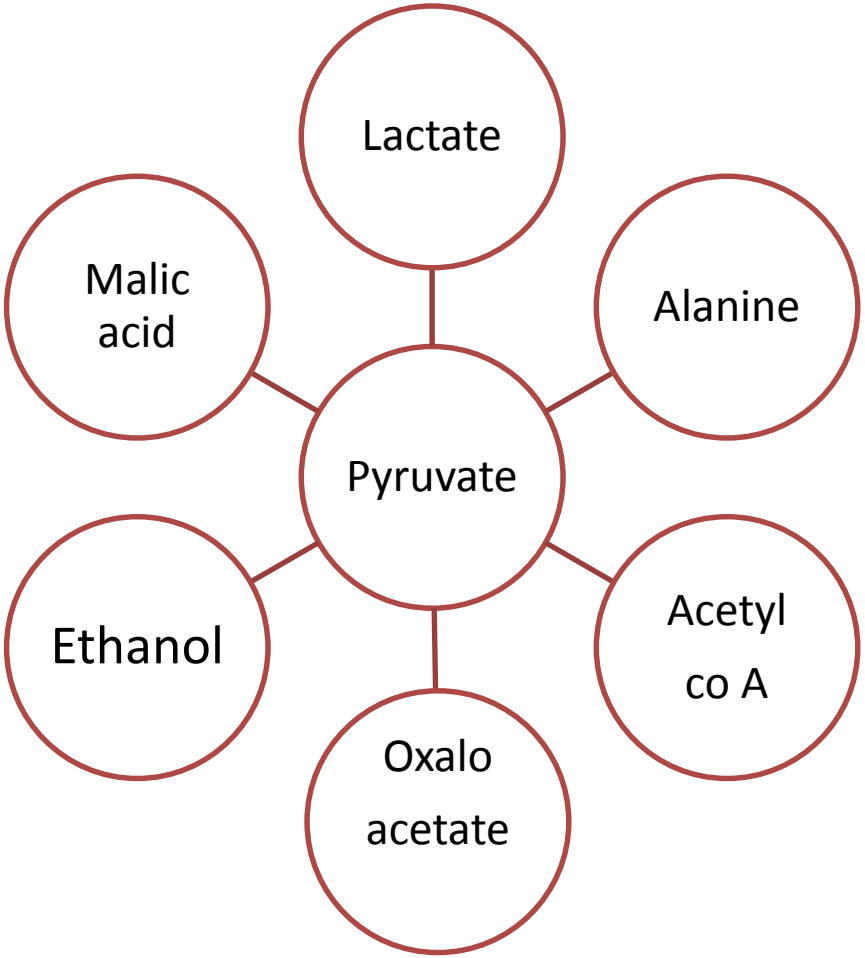
Learning objectives

- To know the different reactions pyruvate undergoes under different cellular conditions and in various cell types
- To know the biological and clinical significance of each of the reactions

Sources of Pyruvate



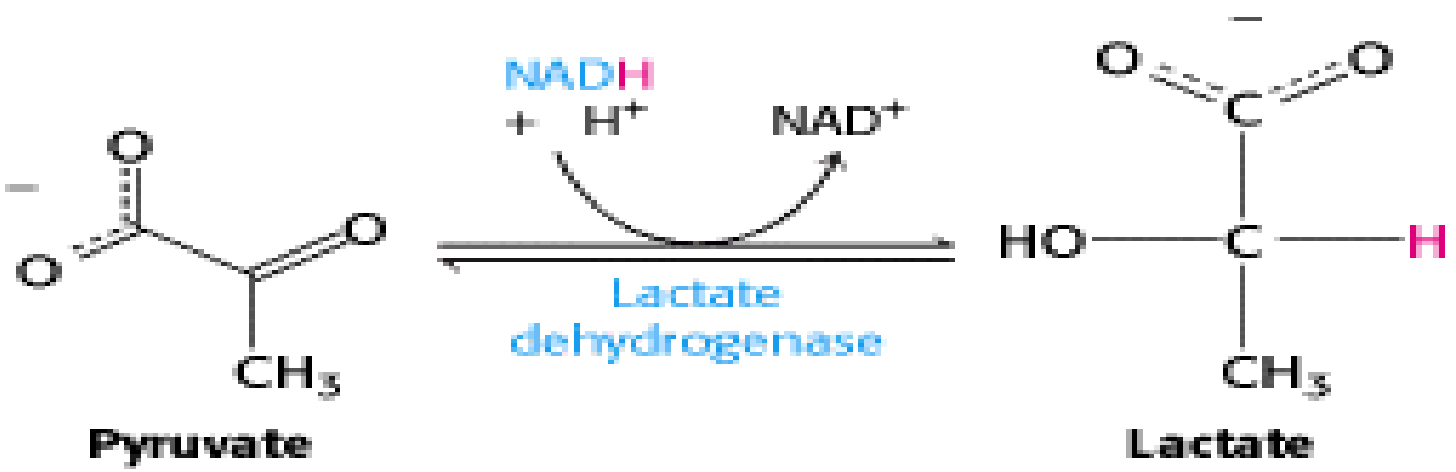
Fate of Pyruvate



1) Pyruvate to lactate conversion

The reduction of pyruvate by NADH to form lactate is catalyzed by ***lactate dehydrogenase***

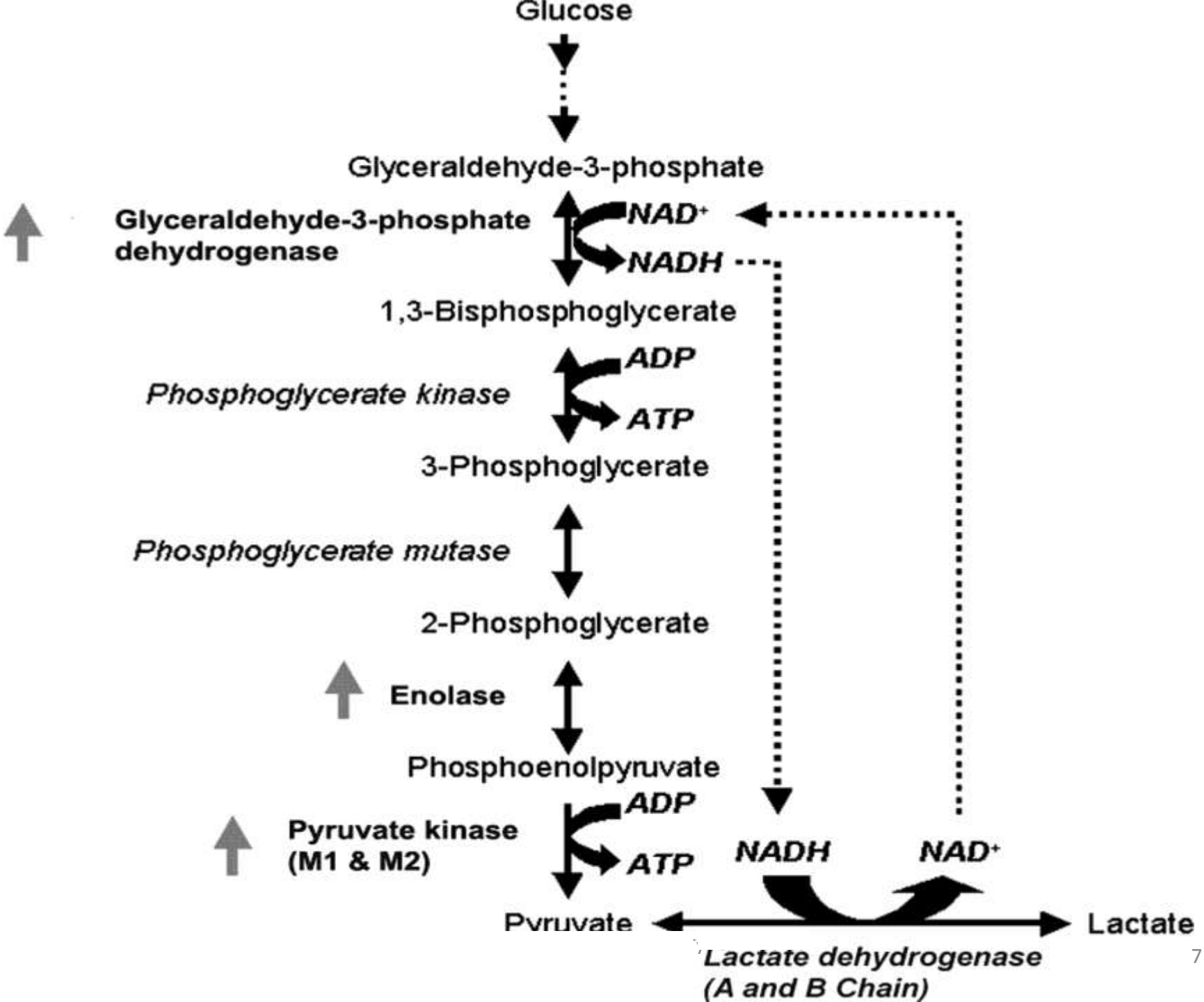
.



The reaction takes place in the cells when the amount of oxygen is limiting, as in muscle during intense activity.

1) Pyruvate to lactate conversion (contd.)

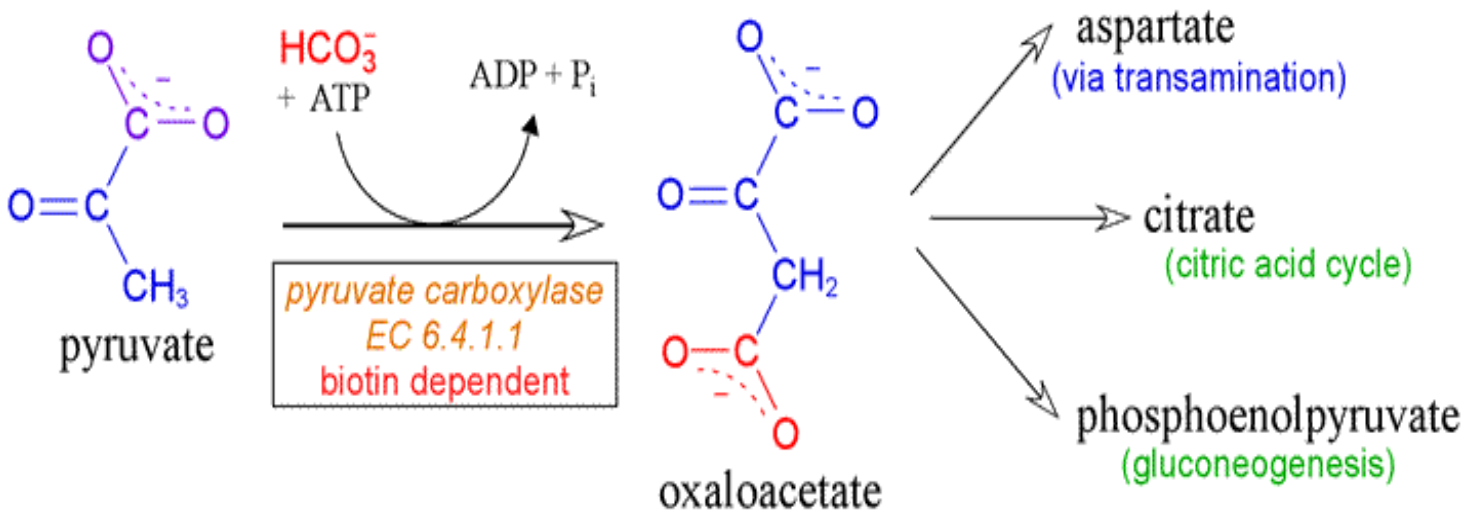
- In the cells lacking mitochondria and under anaerobic conditions, the NADH formed in the oxidation of glyceraldehyde 3-phosphate is consumed in the reduction of pyruvate.
- **The regeneration of NAD^+ in the reduction of pyruvate to lactate sustains the continued operation of glycolysis under anaerobic conditions.**


Glucose
Glyceraldehyde-3-phosphate
Glyceraldehyde-3-phosphate dehydrogenase
NAD+
NADH
1,3-Bisphosphoglycerate
Phosphoglycerate kinase
ADP
ATP
3-Phosphoglycerate
Phosphoglycerate mutase
2-Phosphoglycerate
Enolase
Phosphoenolpyruvate
Pyruvate kinase (M1 & M2)
ADP
ATP
NADH
NAD+
Pyruvate
Lactate
Lactate dehydrogenase (A and B Chain)

2) Pyruvate to Oxaloacetate conversion

- Mitochondrial **pyruvate carboxylase** catalyzes the carboxylation of pyruvate to oxaloacetate, an ATP-requiring reaction in which the vitamin biotin is the coenzyme.
- Biotin binds CO_2 from bicarbonate as carboxybiotin prior to the addition of the CO_2 to pyruvate.

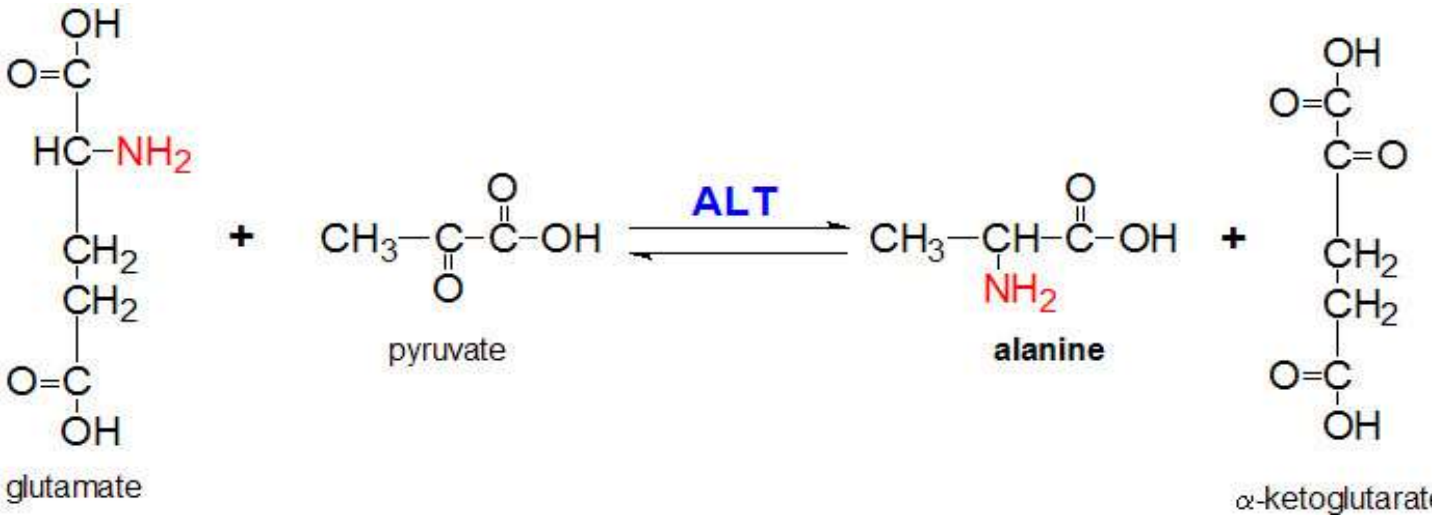
2) Pyruvate to Oxaloacetate conversion (contd.)



- The Oxaloacetate can be subsequently used for the synthesis of Aspartate, phosphoenol pyruvate or
- utilized in the TCA cycle depending upon the need of the cell.

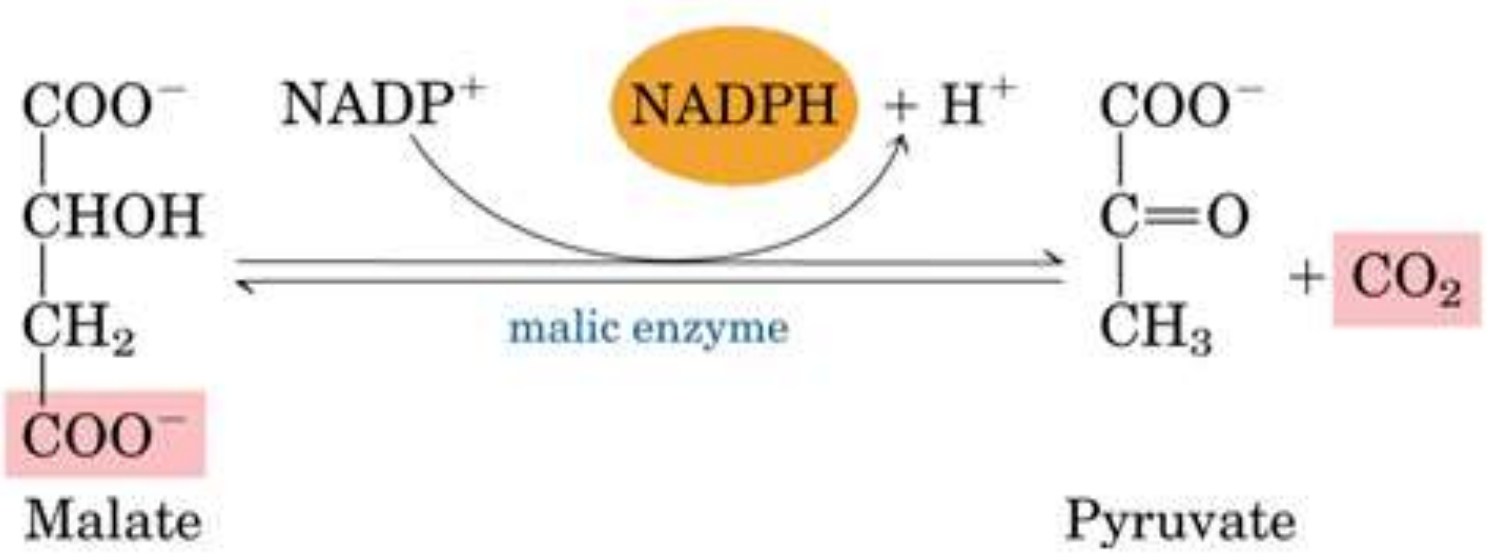
3) Pyruvate to Alanine conversion

- Pyruvate can be transaminated to form Alanine as per the need. The reaction is catalyzed by ALT (Alanine amino transferase).
- This reaction is important for the catabolism and synthesis of non- essential amino acids



4) Pyruvate to Malate conversion

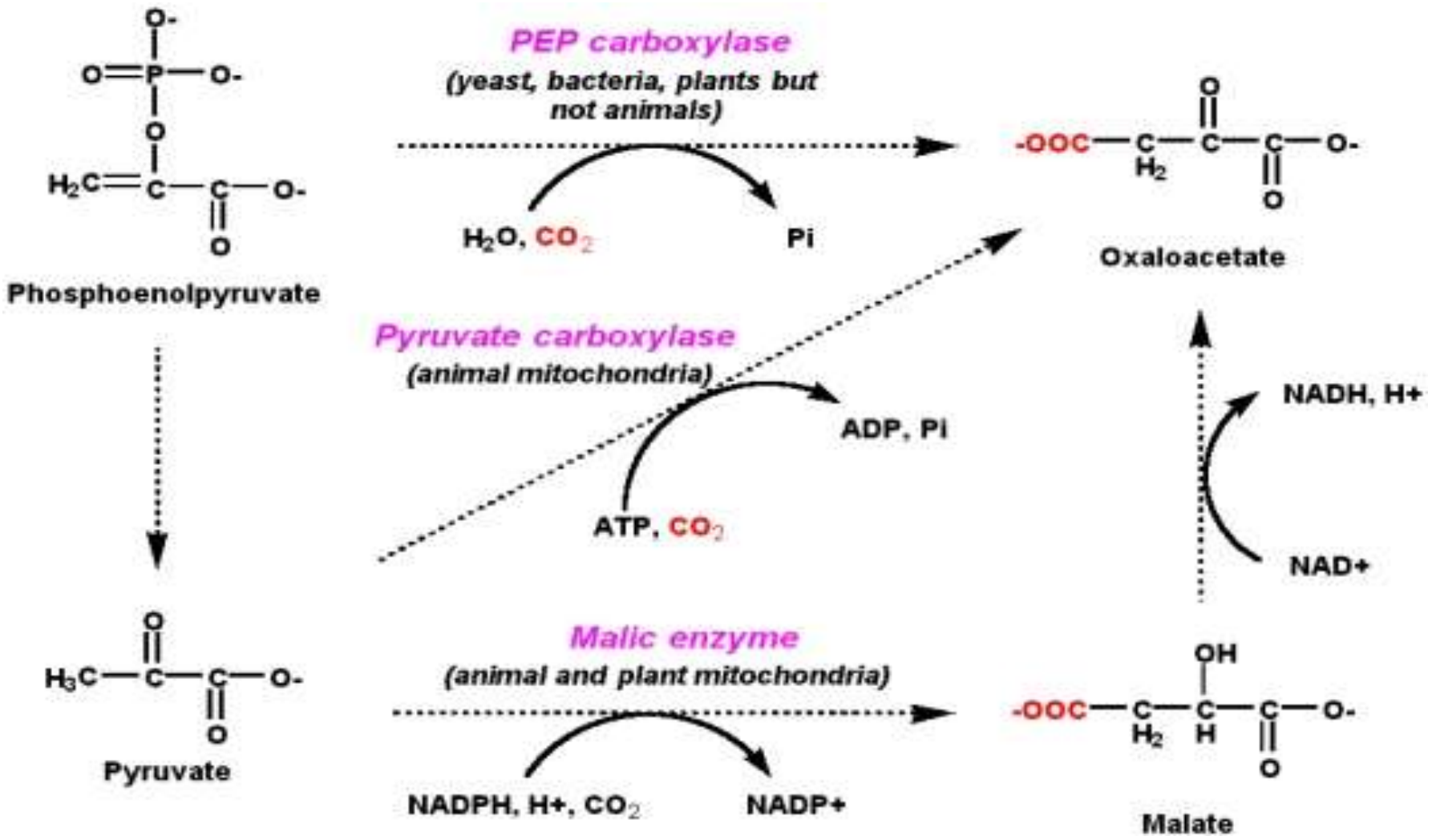
- Pyruvate to malate reaction is a reversible reaction, catalyzed by malate dehydrogenase.
- Cytosolic malate dehydrogenase (Malic enzyme) is an important source for the synthesis of NADPH that can be used for the reductive biosynthesis.



4) Pyruvate to Malate conversion(contd.)

- Pyruvate can be directly converted to oxaloacetate or it is first carboxylated to malate and then decarboxylated to form oxaloacetate.
- These two reactions are called **CO_2 filling up reactions or Anaplerotic reactions**. They provide oxaloacetate when there is sudden influx of Acetyl Co A in the TCA cycle.

4) Pyruvate to Malate conversion

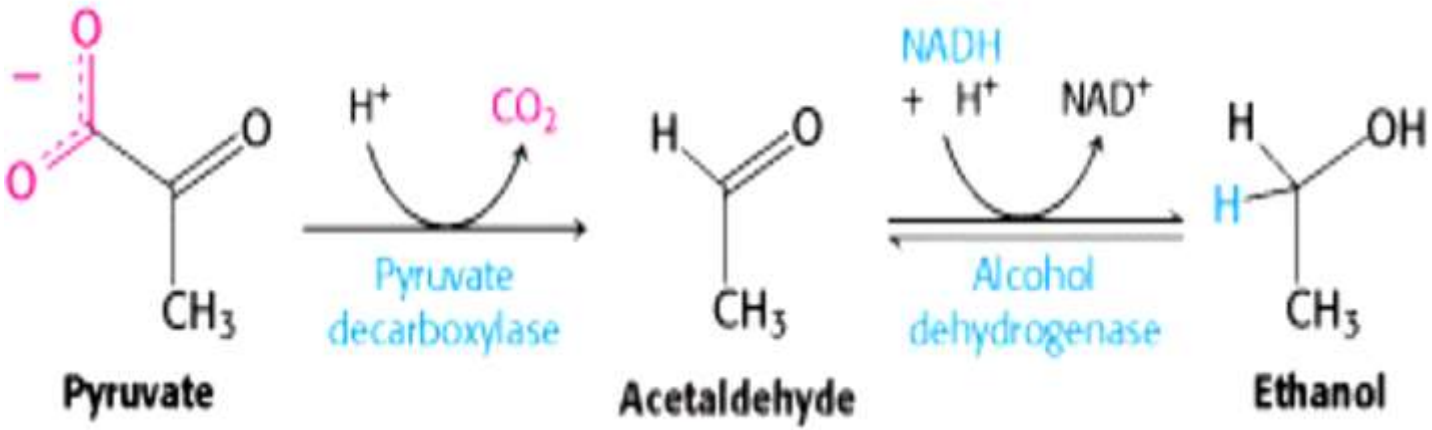


5) Pyruvate to Ethanol conversion

- ***Ethanol*** is formed from pyruvate in yeast and several other microorganisms.
- The first step is the decarboxylation of pyruvate.
- This reaction is catalyzed by ***pyruvate decarboxylase,*** which requires the coenzyme thiamine pyrophosphate (TPP) .

5) Pyruvate to Ethanol conversion (contd.)

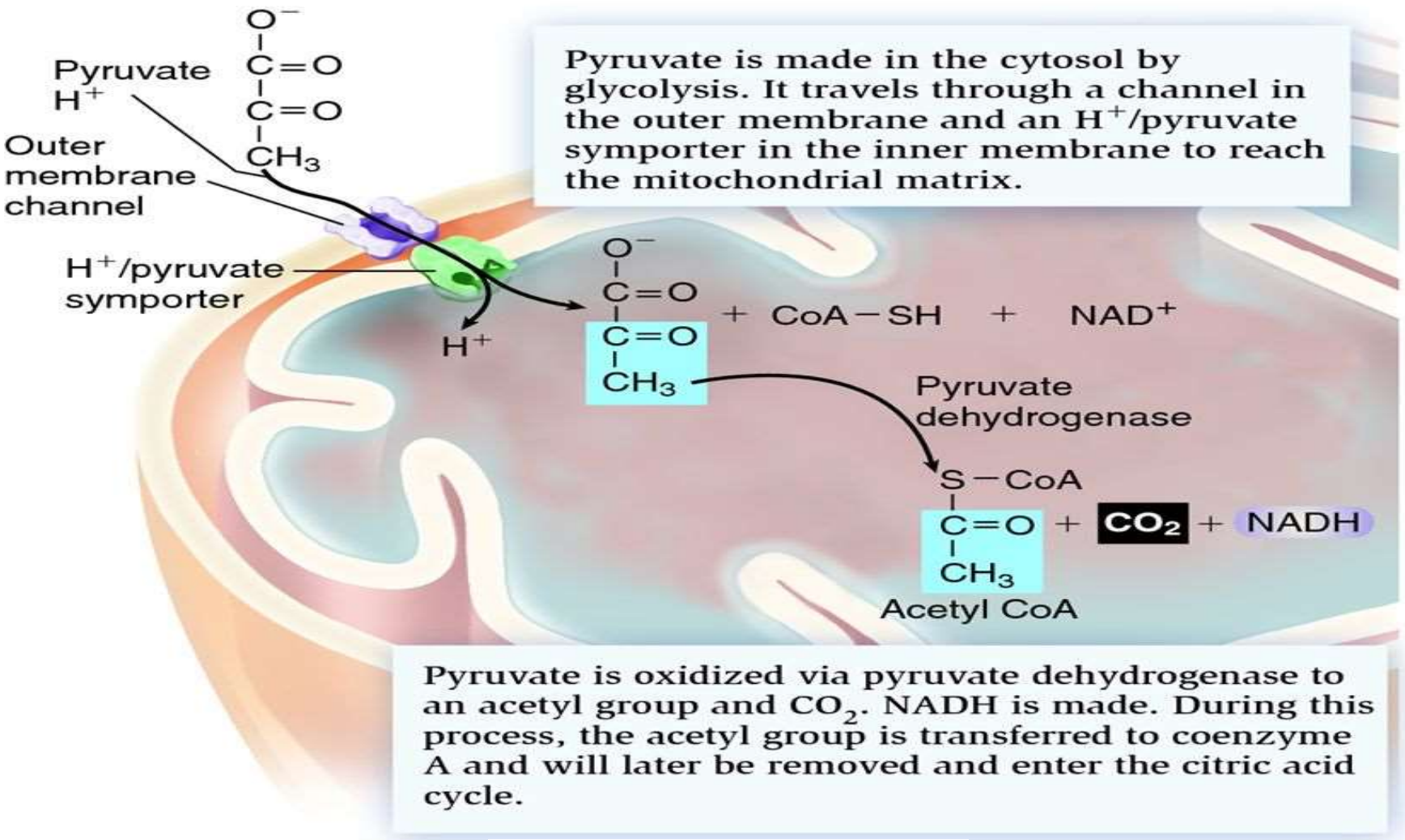
- The second step is the reduction of acetaldehyde to ethanol by NADH, in a reaction catalyzed by *alcohol dehydrogenase*.
- This process regenerates NAD^+.



6) Pyruvate to Acetyl co A conversion

- Under aerobic conditions, pyruvate is transported into mitochondria by a proton symporter.
- In the mitochondrial matrix, pyruvate is oxidatively decarboxylated by the *pyruvate dehydrogenase complex* to form acetyl CoA.
- ***This irreversible reaction is the link between glycolysis and the citric acid cycle.***

6) Pyruvate to Acetyl co A conversion



Components of Pyruvate dehydrogenase complex

1) **Enzymes-** The pyruvate dehydrogenase complex is a large, highly integrated complex of 2 types of enzymes-

A)- Catalytic enzymes

a) Pyruvate dehydrogenase (E1)

b) Dihydrolipoyl transacetylase (E2)

c) Dihydrolipoyl dehydrogenase (E3)

B)- Regulatory Enzymes

a) PDH Kinase

b) PDH Phosphatase

Components of Pyruvate dehydrogenase complex (contd.)

2) Coenzymes of PDH complex

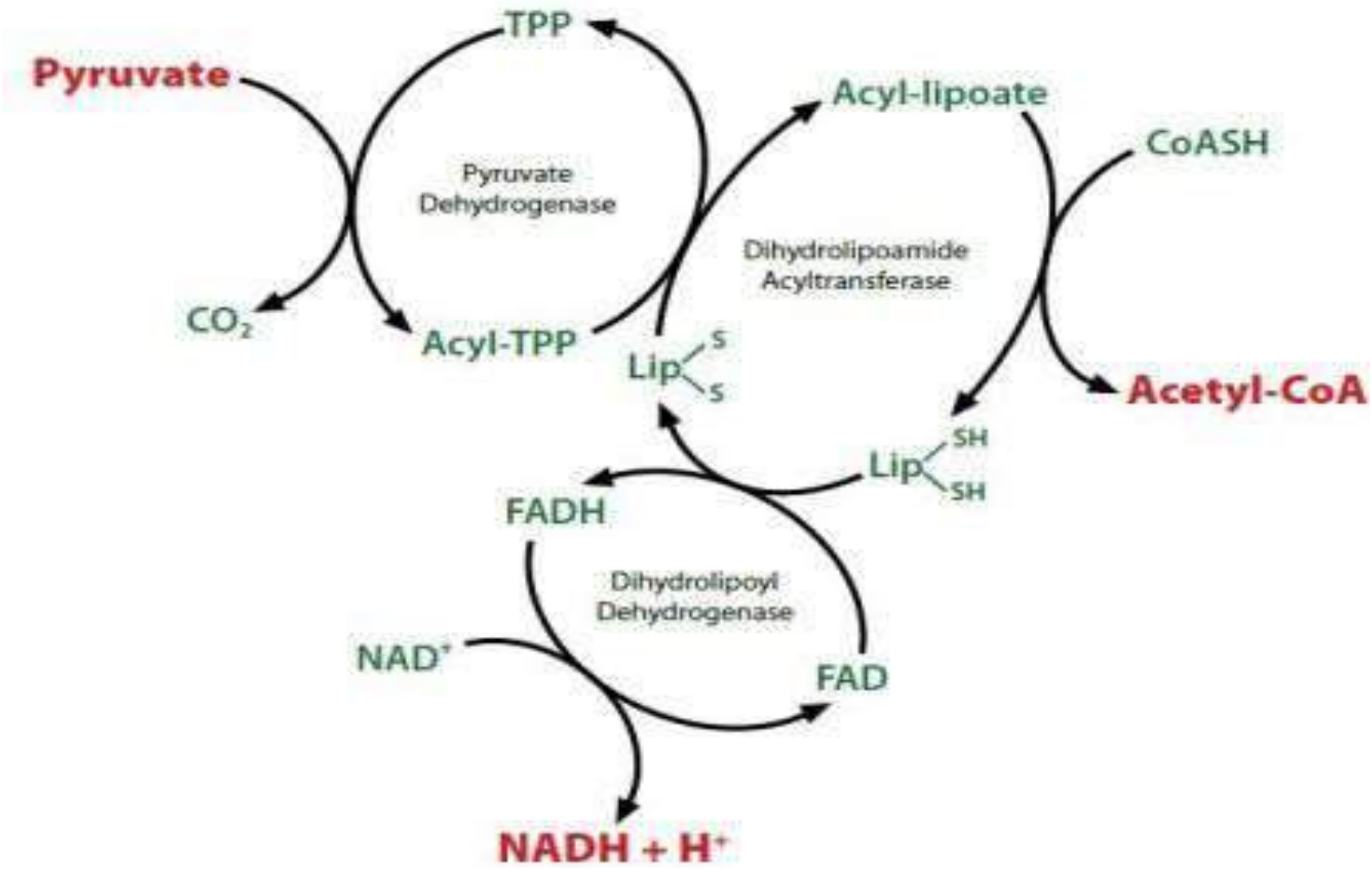
Five coenzymes:

- Thiamine pyrophosphate (TPP),
- Lipoic acid,
- CoASH,
- FAD and
- NAD+ participate in the overall reaction

Reaction catalyzed by PDH Complex

- The conversion of pyruvate into acetyl CoA consists of three steps:
- **Decarboxylation,**
- **Oxidation, and**
- **Transfer of the resultant acetyl group to CoA**
- These steps are coupled to preserve the free energy derived from the decarboxylation step to drive the formation of NADH and acetyl CoA.

Reaction steps



Reaction steps (contd.)

- During the oxidation of pyruvate to CO_2 by pyruvate dehydrogenase,
- the electrons flow from pyruvate to the Lipoamide moiety of dihydrolipoyl transacetylase ,
- then to the FAD cofactor of dihydrolipoyl dehydrogenase and
- finally to reduction of NAD^+ to NADH.

Reaction steps (contd.)

- The acetyl group is linked to coenzyme A (CoASH) in a high energy thioester bond.
- The acetyl-CoA then enters the TCA cycle for complete oxidation to CO_2 and H_2O.

Regulation of PDH complex

- PDH complex is highly regulated by a variety of allosteric effectors and by covalent modification.
- **Allosteric regulation**- Pyruvate dehydrogenase is inhibited by its products,
  - Acetyl-CoA and
  - NADH.

Regulation of PDH Complex (contd.)

Covalent modification- It is also regulated by phosphorylation of three serine residues on the pyruvate dehydrogenase component of the multienzyme complex.

Regulation of PDH Complex (contd.)

PDH exists in two forms-

i) PDH-a form which is active and dephosphorylated form

ii) PDH -b form which is inactive and phosphorylated form

Regulation of PDH complex by covalent modification

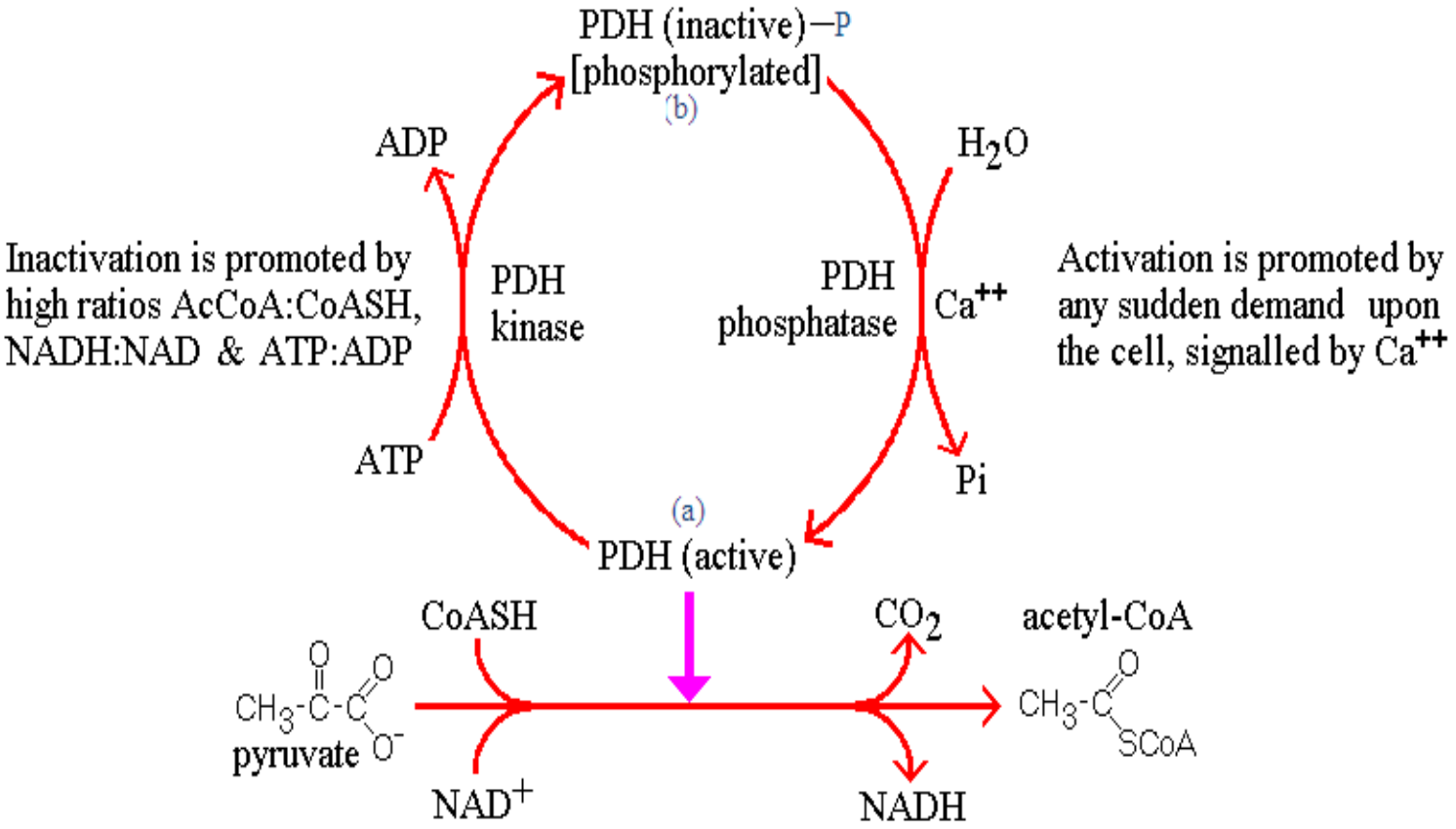
- **PDH kinase**, causes phosphorylation resulting in decreased activity, and
- **PDH phosphatase** causes an increase in activity by dephosphorylation of the enzyme

Regulation of PDH Complex (contd.)

Regulation of PDH Kinase

- **Positive effectors**
- NADH and Acetyl-CoA are powerful positive effectors on PDH kinase,
- The kinase is activated by increases in the [ATP]/[ADP], [Acetyl-CoA]/[CoASH], and [NADH]/[NAD+] ratios.
- the enzyme thus inactivates PDH by converting it to the phosphorylated PDH-b form

Regulation of PDH complex by covalent modification



Regulation of PDH Complex (contd.)

Negative effectors of PDH Kinase

- Pyruvate is a potent negative effector on PDH kinase,
- when pyruvate levels rise, PDH-a (active form) is favored even with high levels of NADH and acetyl-CoA.

Regulation of PDH phosphatase

- Mg2+ and Ca2+ activate the enzyme

Regulation of PDH Complex (conclusion)

- Pyruvate dehydrogenase is inhibited both when there is adequate ATP (and reduced coenzymes for ATP formation) available, and also
- when fatty acids are being oxidized. In fasting, when free fatty acid concentrations increase, there is a decrease in the proportion of the enzyme in the active form, leading to a sparing of carbohydrate.

Energetics of PDH complex

- Two pyruvate molecules are obtained from one glucose molecule through glycolysis.
- Each of the pyruvate yields one NADH, thus there are two NADH molecules to be oxidized through the electron transport chain.

Energetics of PDH complex (contd.)

$$\text{Pyruvate} + \text{CoA} + \text{NAD}^+ \longrightarrow \text{acetyl CoA} + \text{CO}_2 + \text{NADH}$$

- Each of NADH yields 3 ATP molecules,
- thus a total of 6 ATP molecules are produced at the level of PDH complex.

PDH Complex deficiency

- Pyruvate dehydrogenase complex deficiency (PDCD) is a rare disorder of carbohydrate metabolism caused by a deficiency of one or more enzymes in the pyruvate dehydrogenase complex.
- The age of onset and severity of disease depends on the activity level of the PDC enzymes.

Pathophysiology

1) **Energy Deficit**-A deficiency in this enzymatic complex limits the production of citrate.

- Because citrate is the first substrate in the citric acid cycle, the cycle cannot proceed.
- Alternate metabolic pathways are stimulated in an attempt to produce acetyl-CoA; however, an energy deficit remains, especially in the CNS.
- The magnitude of the energy deficit depends on the residual activity of the enzyme.

Pathophysiology (contd.)

2) Neurological deficit

- Severe enzyme deficiencies may lead to congenital brain malformation because of a lack of energy during neural development.
- Underlying neuropathology is not usually observed in individuals whose onset of pyruvate dehydrogenase complex deficiency is in childhood.

Clinical Manifestations

The signs of poor neurological development or degenerative lesions are –

- Poor acquisition or loss of motor milestones,
- poor muscle tone,
- new onset seizures, and
- periods of in-coordination (i.e. ataxia) abnormal eye movements,
- poor response to visual stimuli,
- mental delay, psychomotor delays and
- growth retardation

Laboratory Diagnosis

- High blood lactate and
- High pyruvate levels with or without
- lactic acidemia suggest an inborn error of metabolism at the mitochondrial level.

Treatment of PDH Complex deficiency

- **Cofactor supplementation** with thiamine, carnitine, and Lipoic acid is the standard of care.
- **Ketogenic diets (with restricted carbohydrate intake) have been used to control lactic acidosis** with minimal success.

B R E A K